The following is a reference for the notations used in this course. It may assist you during the video lectures and assignments. Don't worry if you don't understand them yet; they will be covered during the course.

| Notation | Description |
|---|---|
| $A, B, C$ | capital letters represent matrices |
| $u, v, w$ | lowercase letters represent vectors |
| $A$ of size $m \times n$ or $(m \times n)$ | matrix $A$ has $m$ rows and $n$ columns |
| $A^T$ | the transpose of matrix $A$ |
| $v^T$ | the transpose of vector $v$ |
| $A^{-1}$ | the inverse of matrix $A$ |
| $\det(A)$ | the determinant of matrix $A$ |
| $AB$ | matrix multiplication of matrices $A$ and $B$ |
| $u \cdot v; \langle u, v \rangle$ | dot product of vectors $u$ and $v$ |
| $\mathbb{R}$ | the set of real numbers, e.g. $0, -0.642, 2, 3.456$ |
| $\mathbb{R}^2$ | the set of two-dimensional vectors, e.g. $v = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ |
| $\mathbb{R}^n$ | the set of $n$-dimensional vectors |
| $v \in \mathbb{R}^2$ | vector $v$ is an element of $\mathbb{R}^2$ |
| $\|v\|_1$ | L1-norm of a vector |
| $\|v\|_2$; $\|v\|$; $\|\|v\|\|$ | L2-norm of a vector |
| $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3; T(v) = w$ | transformation $T$ of a vector $v \in \mathbb{R}^2$ into the vector $w \in \mathbb{R}^3$ |

✓ Mark As Complete